IE3035 Operations Research

HW5

1. Figure 1 gives the mileage of the feasible links connecting nine offshore natural gas wellheads with an inshore delivery point. Because wellhead 1 is the closest to shore, it is equipped with sufficient pumping and storage capacity to pump the output of the remaining eight wells to the delivery point. Determine the minimum pipeline network that links the wellheads to the delivery point.

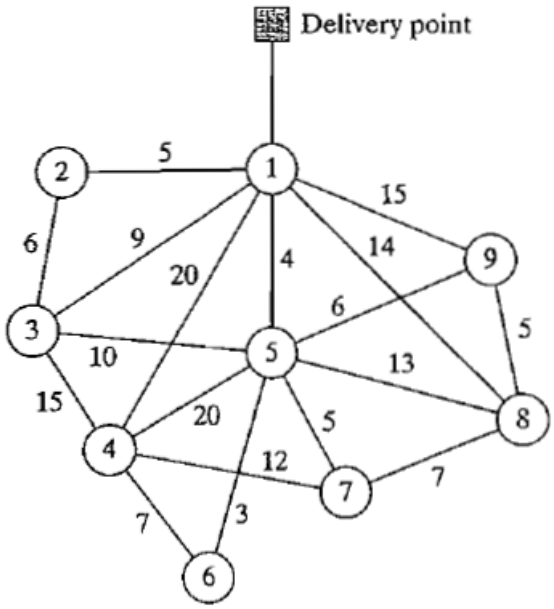


Figure 1

2. Figure 2 provides the communication network between two stations, 1 and 7. The probability that a link in the network will operate without failure is shown on each arc. Messages are sent from station 1 to station 7, and the objective is to determine the route that will maximize the probability of a successful transmission. Formulate the situation as a shortest-route model and determine the optimum solution.

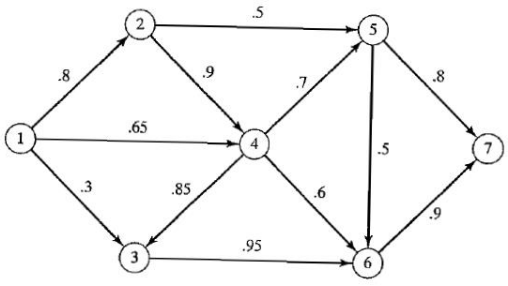


Figure 2

3. Jobco is planning to produce at least 2000 widgets on three machines. The minimum lot size on any machine is 500 widgets. The following table gives the pertinent data of the situation.

Machine	Cost	Production cost/unit	Capacity (units)
1	300	2	650
2	100	10	850
3	200	5	1250

Formulate the problem as an ILP, and find the optimum solution.

4. Develop the B&B tree for each of the following problems. For convenience, always select x_1 as the branching variable at node 0.

Maximize $z = 2x_1 + 3x_2$

subject to

$$5x_1 + 7x_2 \leq 35$$

$$4x_1 + 9x_2 \leq 36$$

$x_1, x_2 \geq 0$ and integer